

ACTIVITY I

DATA VISUALIZATION AND DASHBOARD DEPLOYMENT

Learning objectives:

At the end of this activity the student will be able to:

- a. *Load and explore datasets using Python and perform basic descriptive visualizations.*
- b. *Create interactive data visualizations in Google Colab and prepare them for deployment.*
- c. *Develop a simple analytical dashboard in Streamlit to display key metrics and insights.*
- d. *Integrate Colab, GitHub, and Streamlit Cloud in a complete data-analysis workflow.*

A university aims to gain deeper insight into its student admission, enrollment, retention, and satisfaction processes in order to support data-driven decision-making. The institution has provided a dataset (university_student_data.csv) containing information about student applications, enrollment, retention, and satisfaction across several academic years and departments.

The goal is to design and implement an analytical dashboard that summarizes and visualizes this information clearly and interactively.

Instructions

1. In a Google Colab notebook, load the dataset and perform an initial exploratory analysis. Clearly document the available columns and explain the meaning of each one in the context of the study.
2. Create visualizations to explore and summarize the following indicators of the dataset. Use different types of plots depending on the purpose of the analysis, for example: plot() to visualize changes over time, bar charts to compare categories and sns.lineplot() to show trends between variables.
 - Retention rate trends over time.
 - Student satisfaction scores by year.
 - Comparison between Spring and Fall terms.

Note: To generate these visualizations, use groupby() to select and aggregate the relevant columns from the dataset, and apply matplotlib or seaborn to create the corresponding plots.

3. Once you have generated your visualizations, build an interactive dashboard prototype using [Streamlit Cloud](#). Your app should:
 - Include interactive filters (year, department, or term).
 - Contain at least three different visualization types (for example: bar chart, line chart, pie/donut, KPI cards).
 - Ensure that all graphs and indicators update dynamically according to the selected filters.
4. Link your Colab notebook to a GitHub repository where you will store all files (app.py, requirements.txt, and README.md).
5. Deploy your dashboard on Streamlit Cloud directly from your GitHub repository. Verify that the deployed version works correctly before submission.
6. In a Word document, summarize the main findings from your analysis. Highlight

the most relevant trends observed in the dashboard and suggest at least one actionable insight that could help the university improve its admission or retention processes.

Deliverables

- A Word or PDF document (maximum 1 page) that includes: the GitHub repository link and the deployed Streamlit dashboard link.
- The GitHub repository must contain:
 - app.py (Streamlit dashboard)
 - requirements.txt
 - README.md describing purpose and instructions.
 - All team members' names must appear in the GitHub repository and the deployment.

The activity must be completed in **groups of at least 2 and at most 4 participants.**

Best of luck!

*“The three chief virtues of a programmer are:
Laziness, Impatience and Hubris.”
Larry Wall*